Bibimcad

How to install the Phase 1 add-in in Autodesk Revit 2023

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Bibimcad is a C# Revit add-in. This computer does not have Revit, so the DLL has not been built yet. The person who installs it needs Revit 2023 and Visual Studio 2022 on their own PC.

1. What that PC needs

- Autodesk Revit 2023
- Visual Studio 2022
- .NET Framework 4.8, which Visual Studio uses to build the project
- The Bibimcad source folder, including the solution file

> Do not copy `RevitAPI.dll` or `RevitAPIUI.dll` into the project. Revit already has those files. The project reads them from the Revit 2023 install folder.

2. Files to bring across

Copy the whole source folder, then leave out `src\Bibimcad.Addin\obj\`. That folder is only a failed build from a PC without Revit.

Keep	Why
`Bibimcad.sln`	Open this in Visual Studio.
`src\Bibimcad.Addin\`	Project file and C# source.
`data\project_state.json`	Template for the central project file.
`installer\Bibimcad.addin`	Manifest Revit reads at startup.
`docs\architecture.md`	How the add-in is structured.

3. Build the DLL

1. Open `Bibimcad.sln` in Visual Studio 2022.
2. Confirm the project target is **.NET Framework 4.8**.
3. If `RevitAPI` or `RevitAPIUI` shows as missing, add a reference to both DLLs in the Revit 2023 install folder. The usual folder is `C:\Program Files\Autodesk\Revit 2023`. Set **Copy Local** to **False** on both references.
4. Restore NuGet packages. The project uses **Newtonsoft.Json 13.0.3**.
5. Build **Debug**.

A successful build writes these two files next to each other:

```
src\Bibimcad.Addin\bin\Debug\Bibimcad.Addin.dll
src\Bibimcad.Addin\bin\Debug\Newtonsoft.Json.dll
```

If Revit 2023 is installed in another folder, the reference path can be overridden:

```
dotnet build -p:RevitInstallDir="D:\Path\To\Revit 2023"
```

4. Put the project file where Bibimcad reads it

The running add-in does not read the JSON file inside the source folder. Copy the template to this exact path, and create the folders if they are missing:

```
C:\Bibimcad\data\project_state.json
```

The source file to copy is `data\project_state.json`. Logs are written later to `C:\Bibimcad\logs\Bibimcad.log`.

5. Install the manifest

1. Create this folder if it does not exist:

```
%APPDATA%\Autodesk\Revit\Addins\2023\
```

On a normal PC that is:

```
C:\Users\<user name>\AppData\Roaming\Autodesk\Revit\Addins\2023\
```

2. Copy `installer\Bibimcad.addin` into that folder. The file name stays `Bibimcad.addin`.
3. Open the copied file in Notepad. Replace the assembly placeholder with the full path of the DLL from the build. Keep the class name as it is.

```
<Assembly>FULL_DLL_PATH_PLACEHOLDER</Assembly>
```

Example, using the path on the machine where the project was created:

```
<Assembly>F:\Cursor_Projects\BibimcadCsharp\src\Bibimcad.Addin\bin\Debug\Bibimcad.Addin.dll</Assembly>
```

Use the real folder on the Revit PC if the project was copied somewhere else. The line `<FullClassName>Bibimcad.App</FullClassName>` stays unchanged.

> `Newtonsoft.Json.dll` must remain in the same folder as `Bibimcad.Addin.dll`. Revit loads it from there. Pointing the manifest at the DLL in `bin\Debug` is enough, as long as both files stay together.

6. Start Revit 2023

1. Close Revit if it is already open, then start Revit 2023.

2. If Revit asks whether to load Bibimcad, choose **Always Load**.
3. Look for a ribbon tab named **Bibimcad**.
4. On that tab, the panel **K1 Workflow** has two buttons: **01 Inspect** and **02 Coordinates**.

7. Check that Phase 1 works

1. Open a Revit project that contains a linked CAD file named `K1.dwg`.
2. Run **01 Inspect**. The dialog should show Bibimcad, the project name **K1 BIM Conversion**, Revit 2023, LOD 200, the DXF path, the approved unit **Millimeters**, and that unit calibration is approved.
3. Run **02 Coordinates**. If exactly one linked CAD file matches `K1.dwg`, Bibimcad selects it. The dialog shows the origin in millimeters, rotation, scale, and whether the 6000 mm distance check passed. The same transform is saved back into `C:\Bibimcad\data\project_state.json`. Origin values inside that file stay in feet.

> Phase 1 does not create walls or any other Revit geometry. It does not read the DXF file yet.

8. If the tab does not appear

- The `<Assembly>` path does not point at the real `Bibimcad.Addin.dll`.
- `Newtonsoft.Json.dll` is missing from the same folder as the add-in DLL.
- The manifest is in the Addins folder for a different Revit year. It must be the **2023** folder.
- Revit was left open while the manifest was copied. Restart Revit.
- The Debug build did not finish, so the DLL does not exist yet.

After a command runs, messages are also written to `C:\Bibimcad\logs\Bibimcad.log`.

Bibimcad Phase 1 foundation. Build and test this in Revit 2023 before starting Phase 2.